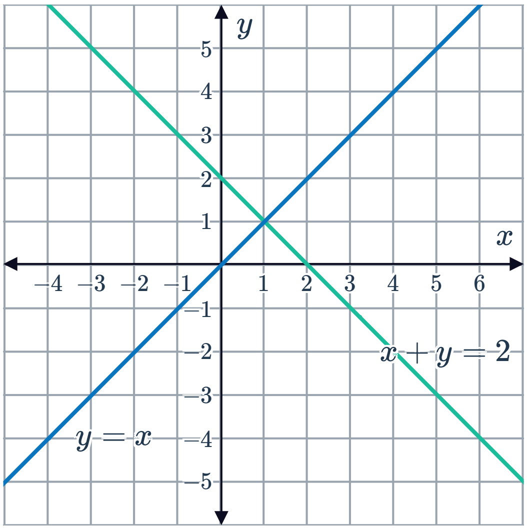
Areas under multiple curves

1

a

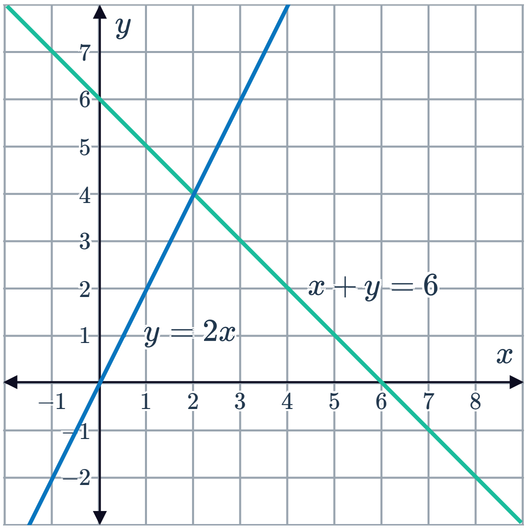
i



ii 1 unit^2

b

i

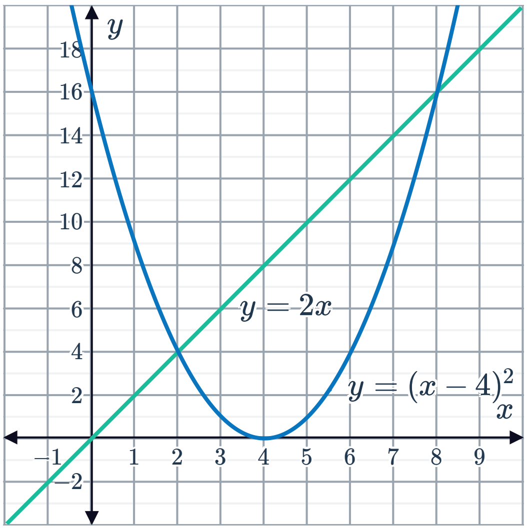


ii 12 units^2

2

a

i

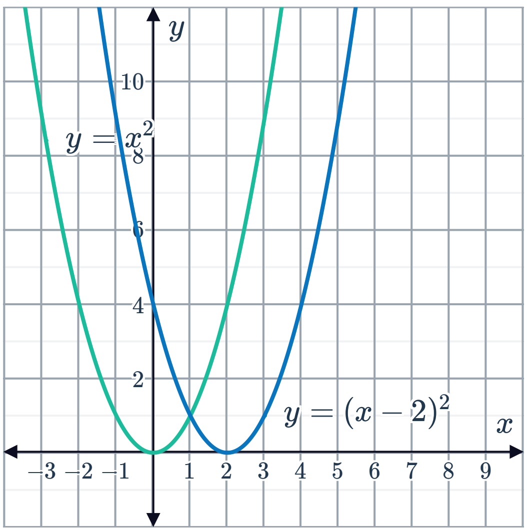


ii $x = 2, 8$

iii $\frac{20}{3}$ units^2

b

i



ii $x = 1$

iii $\frac{2}{3}$ units^2

3

a $\frac{14}{3}$ units^2 **b** $\frac{103}{12}$ units^2 **c** $\frac{44}{3}$ units^2 **d** $\frac{11}{24}$ units^2

4 $\frac{7}{6}$ units^2

5 $\frac{25}{3}$ units^2

6 $\frac{3}{2} - \frac{1}{2}e^{-4} - e^{-2}$ units^2

7 $\frac{89}{3}$ units^2

8 $\frac{50}{3}$ units^2

9

a 8 units^2 **b** 16 units^2 **c** $2:1$

10 $k=2$

11 $4-\pi$ units^2

12

a $x=9$ **b** $\left(3e^{\frac{3}{2}}+6\right)$ units^2

13

a $\left(e^3-1\right)$ units^2 **b** $y=e^3x-2e^3$ **c** $\left(\frac{1}{2}e^3-1\right)$ units^2

Areas between curves

14 30 units^2

15 $\frac{28}{3}$ units^2

16 $\frac{4}{15}$ units^2

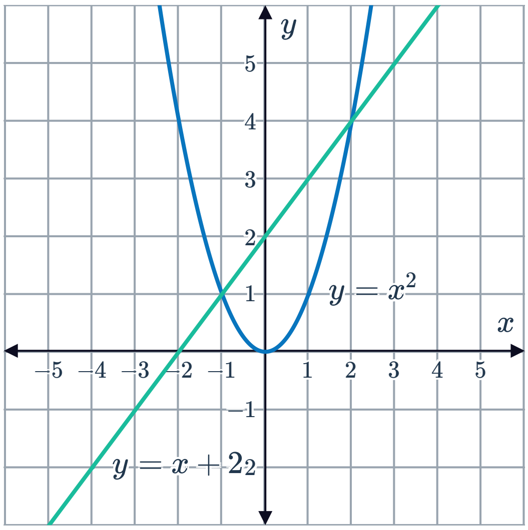
17

a $x=0,\ 4,\ 9$ **b** $\frac{2521}{12}$ units^2 **c** $\frac{100}{3}$ units^2

18

a

i

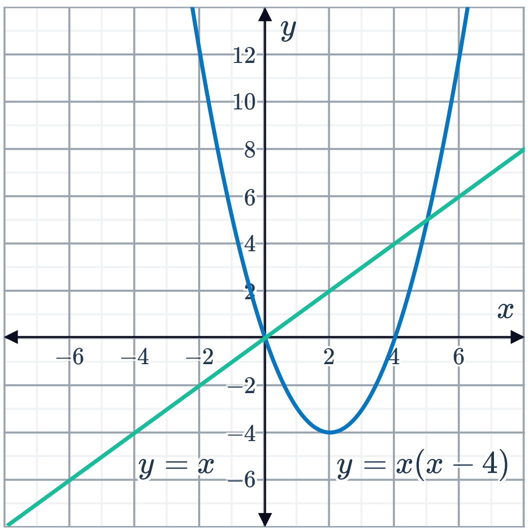


ii $x = -1, 2$

iii $\frac{9}{2}$ units^2

b

i

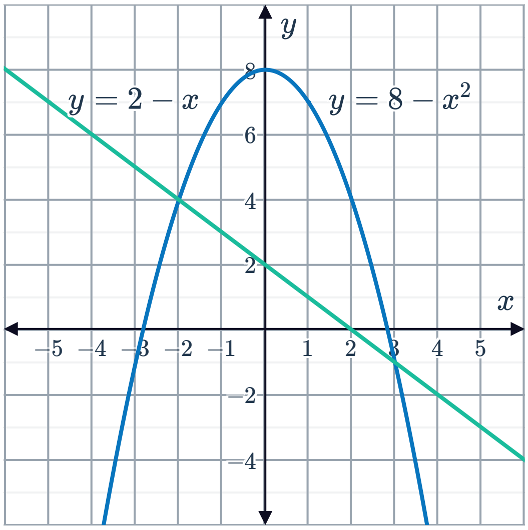


ii $x = 0, 5$

iii $\frac{125}{6}$ units^2

c

i

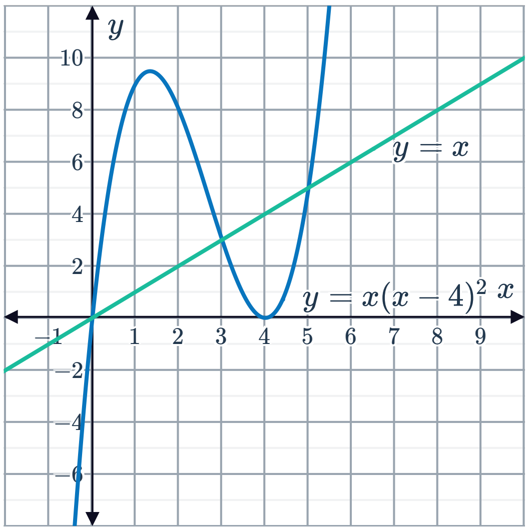


ii $x = -2, 3$

iii $\frac{125}{6}$ units^2

d

i

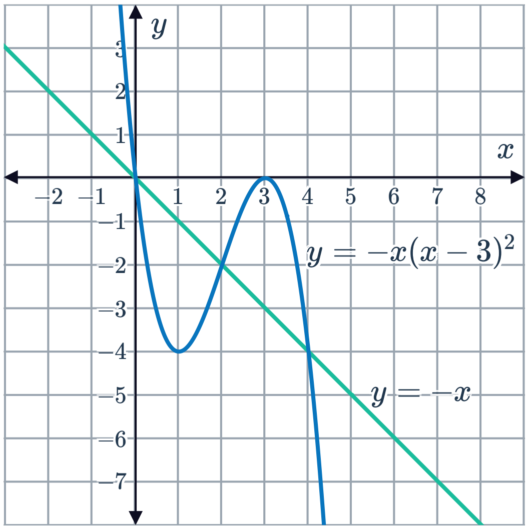


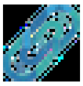
ii $x = 0, 3, 5$

iii $\frac{253}{12}$ units^2

e

i





ii $x = 0,\ 2,\ 4$

iii 8 units^2

19

a

i $x = -4,\ 6$

ii $\frac{1000}{3}$ units^2

b

i $x = 4,\ 5,\ 6$

ii $\frac{1}{2}$ units^2

c

i $x = -1,\ 3,\ 4$

ii $\frac{131}{4}$ units^2

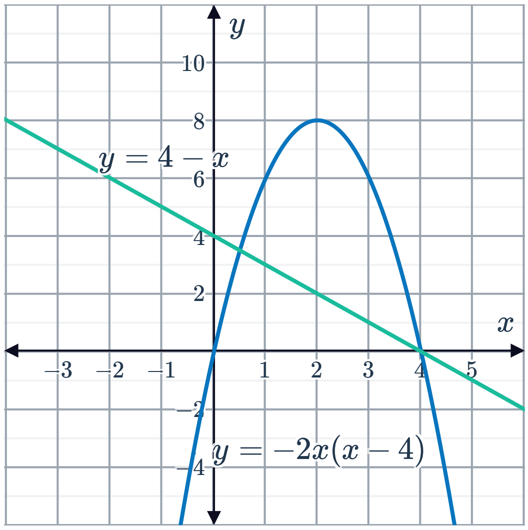
d

i $x = 5,\ -3$

ii $\frac{256}{3}$ units^2

20

a



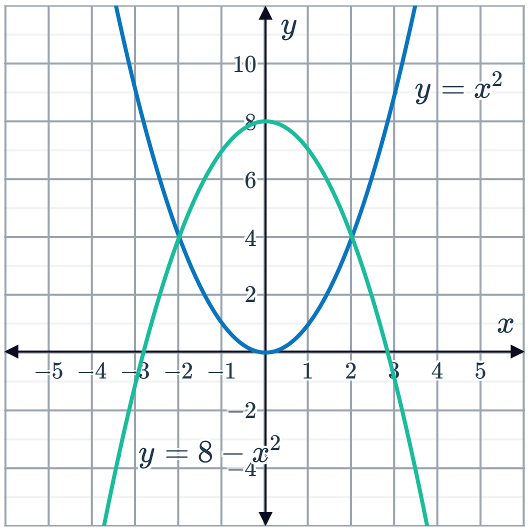
b $x = \frac{1}{2}, 4$

c $\frac{343}{24}$ units^2

d $\frac{23}{24}$ units^2

21

a



b $x = -2, 2$

c $\frac{64}{3}$ units^2

d 8.8 units^2

22

a

i Yes **ii** Yes **iii** No **iv** Yes

v No **vi** No

b 4 units^2

c $2 : 1$

23 $\frac{2}{\sqrt{2}} - 1$ units^2

$\frac{1}{4}$ units^2

25 $\frac{1}{4}\left(\sqrt{2}-1\right)$ units^2

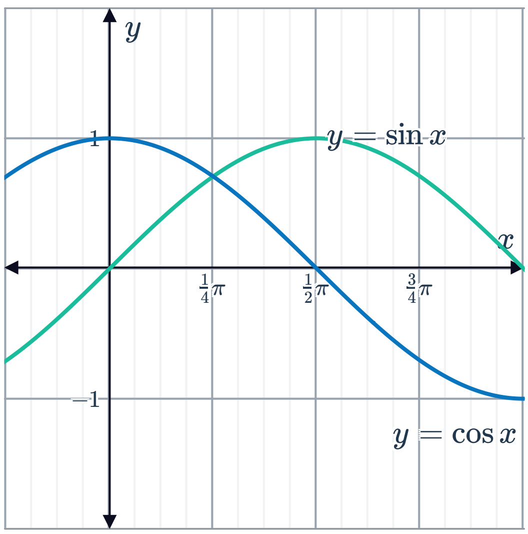
26 11.89 units^2

27 $4\sqrt{2}+4$ units^2

28 $\frac{10\pi^2}{9}+3$ units^2

29

a

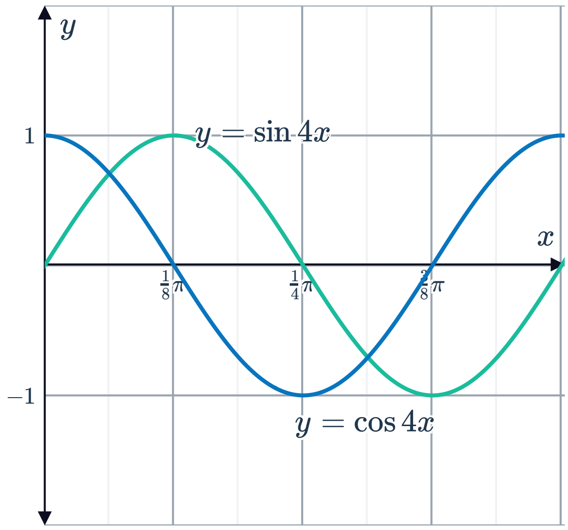


b $2\sqrt{2}$ units^2

30

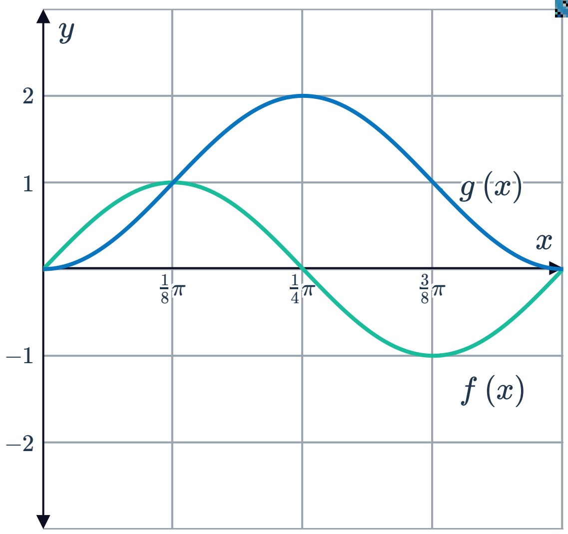
a $x = \frac{\pi}{16}, \frac{5\pi}{16}$

b



c $\frac{\sqrt{2}}{2}$ units^2

31 **a**



b $0, \frac{\pi}{8}, \frac{\pi}{2}$

c $\frac{\pi}{2}$

d $\frac{\pi}{4} + 1 \text{ units}^2$

32

a 4 units^2

b $e^3 + e^{-3} - 2 \text{ units}^2$

c $\frac{2}{5}e^{10} + \frac{2}{5}e^{-10} - \frac{4}{5} \text{ units}^2$

d $3 + e^{-2} \text{ units}^2$

33

a $x = 3$

b $2e^3 - 17 \text{ units}^2$

34

a $x = 0, 1$

b $\frac{1}{3} \text{ units}^2$

c

n	2	3	5	8
Area	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{7}{9}$

d The area between the curves $f(x) = x^n$ and $g(x) = \sqrt[n]{x}$ for positive integer values of n is $\frac{n-1}{n+1}$.

.

e No solution provided.

35

a -22

b 34

c 13 units^2

d 5 units^2

e 29 units^2

36

a 16

b -10

c 1 unit^2

d 20

37

a 18 units^2

b $\frac{22}{3} \text{ units}^2$

c $a = \frac{182}{75}$